

9DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2024

Course: B. Tech

Semester : III

Branch : Electronics Engineering/ Electronics and Telecommunication Engineering

Subject Code & Name: Electronic Devices & Circuits/ BTEXC303_Y18

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
What does JFET stand for?	CO1	1
a) Junction Field-Effect Transistor.		
b) Junction Frequency-Effect Transistor.		
c) Junction Forward-Effect Transistor.		
d) Junction Feedback-Effect Transistor.		
2 What is the pinch-off voltage in a JFET?	CO1	1
a) The voltage at which the channel is fully closed and no current flows.		
b) The voltage at which the JFET starts conducting.		
c) The voltage at which the JFET breaks down.		
d) The voltage at which the JFET enters the saturation region.		
3 What is the purpose of biasing in a JFET?	CO2	1
a) To set the operating point in the desired region.		
b) To increase the gain of the JFET.		
c) To reduce the input impedance.		
d) To decrease the output impedance.		
4 What is the primary function of the gate in a MOSFET?	CO3	1
a) To control the flow of current between the source and drain.		
b) To provide power to the MOSFET.		
c) To act as a heat sink.		
d) To connect the MOSFET to the ground.		

5	What is the threshold voltage in an E-MOSFET?	CO3	1
	a) The minimum gate-to-source voltage required to form a conductive channel. b) The maximum drain-to-source voltage. c) The voltage at which the MOSFET breaks down. d) The voltage at which the MOSFET enters the saturation region.		
6	In which region does a MOSFET operate when used as a switch?	CO3	1
	a) Cut-off region. b) Ohmic region. c) Saturation region. d) Breakdown region.		
7	Which of the following amplifiers is designed to amplify weak signals with minimal addition of noise?	CO4	1
	a) Power Amplifiers b) Low Noise Amplifiers c) Voltage Amplifiers d) Operational Amplifiers		
8	What is the primary purpose of a power amplifier?	CO4	1
	a) To amplify weak signals b) To drive loudspeakers or high-power loads c) To filter noise d) To provide high-frequency gain		
9	Which of the following oscillators is most commonly used for audio frequency generation?	CO4	1
	a) Hartley Oscillator b) Colpitts Oscillator c) Wein Bridge Oscillator d) Crystal Oscillator		
10	The Barkhausen criterion is used in the analysis of which type of circuits?	CO4	1
	a) Amplifiers b) Oscillators c) Filters d) Attenuators		
11	What is the main feature of a monostable multivibrator?	CO5	1
	a) It has two stable states and produces a square wave. b) It has one stable state and produces a single pulse when triggered. c) It has no stable states. d) It operates with a fixed frequency.		

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|----|---|-----|---|
| 12 | A Switch Mode Power Supply (SMPS) is more efficient than a linear power supply because: | CO5 | 1 |
| | a) It generates no heat. | | |
| | b) It uses a low-frequency transformer. | | |
| | c) It switches on and off, reducing power loss. | | |
| | d) It operates without feedback. | | |

Q. 2 Solve the following.

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|----|---|-----|---|
| A) | Determine I_{DQ} , V_{GSQ} , V_D , V_S , V_{DS} and V_{DG} for the network of figure 1. | CO1 | 6 |
|----|---|-----|---|

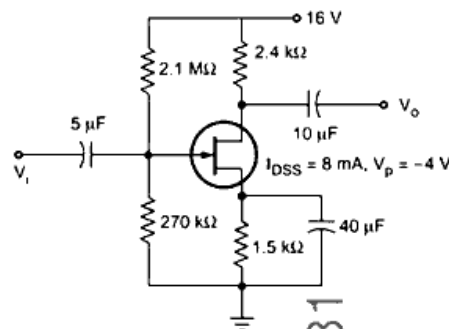


Figure 1

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| B) | Compare BJT, FET and MOSFET with different parameters. | CO1 | 6 |
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Q.3 Solve the following. 12

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|----|---|---------|---|
| A) | Explain the drain characteristics and transfer characteristics of N-channel depletion type MOSFET. | CO1 | 6 |
| B) | Determine the d.c bias point for the EMOSFET circuit shown in figure 2. Assume $k=0.4\text{mA/V}^2$, $V_T=3\text{V}$. | CO1/CO2 | 6 |

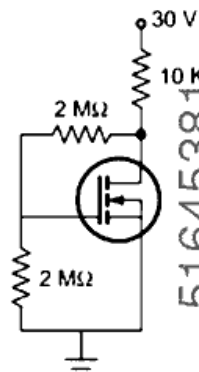


Figure 2

Q. 4 Solve Any Two of the following. 12

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|----|--|-----|---|
| A) | With a neat circuit diagram explain the working of a MOSFET as a switch. | CO3 | 6 |
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| B) Explain the classification of power amplifiers in detail. | CO3 | 6 |
| C) What is a feedback amplifier? Explain the types of feedback amplifiers. | CO3 | 6 |

Q.5 Solve Any Two of the following. 12

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|--|---------|---|
| A) With a neat circuit diagram derive the expression of frequency of oscillations for a Colpitts oscillator. | CO3/CO4 | 6 |
| B) State and explain Barkhausen criterion of Oscillations. | CO4 | 6 |
| C) Explain the concept of frequency stability of Oscillators. | CO4 | 6 |

Q. 6 Solve Any Two of the following. 12

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|--|-----|---|
| A) What is a multivibrator. Explain the operation of monostable multivibrator using IC555 timer. | CO4 | 6 |
| B) With a neat block diagram explain the IC555 timer. | CO5 | 6 |
| C) Draw the block diagram of SMPS and also describe its operation. | CO5 | 6 |

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